THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 16, 2022 09:30 pm – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) With the aid of appropriate diagram(s), explain the mechanism of "traditional" and "design and build" procurement systems. [15 marks]

   b) Discuss the "building own operate transfer (BOOT)" system with suitable the examples. [10 marks]

2. a) An investor is planning to build a 100-unit service apartment. The usable floor area of this service apartment will be 6,000m^2 while the circulation area will take up 20% of the total floor space. The construction cost is estimated to be \$10,000/m^2 and the building will take 2 years to construct. At a monthly rent of \$28,000/unit, the service apartment will attract 90% occupancy during its first three years of operation before attaining a full occupancy rate from the fourth year onward. The initial yield for this type of development would be 6%. With the cost of disposal being \$1,000,000 and a short-term finance of 12% per annum. By considering the developer's profit (15%), professional fee (13%), contingency (5%) and acquisition costs (4%), calculate the maximum site value based on the above parameters and your engineering judgement. [15 marks]

   b) Briefly describe the major advantages and disadvantages of residual method, and under what circumstances should discount cash flow methods be used for investment appraisal. [5 marks]

   c) With the aid of a sensitivity diagram, discuss the uncertainty of financial viability for different variables based on internal rate of return (IRR). [5 marks]

3. a) With the aid of diagrams, briefly describe and distinguish the outline and master construction programmes. [15 marks]

   b) Define activity logic. With appropriate examples, discuss different types of activity relationship. [10 marks]

4. a) Discuss whether it is ethical for contractors to "*forming ring*" or submit a "*cover price*" and how should consultant civil engineers deal with these situations competently? [7 marks]

   b) Compute the unit rate for a 100 mm thick in-situ reinforced concrete (1:2:4, 40 mm aggregate) wall based on the basic cost data as shown below:

Material		
Type		*Material Price*
Cement		HK$ 1000.00/tonne
Sand		HK$ 170.00/tonne
Aggregate		HK$ 160.00/tonne
Labour		
Type		*Wage*
Concrete vibrator		HK$ 170.00/hour
All other workers		HK$ 150.00/hour
Location	**Labour Constants**	
	Man hour	*Gang hour*
< 100 mm	9.36	2.34
100 – 150 mm	8.00	2.00
Plant		
Type		*Hiring Charge*
Mixer (output = 4 m^3/ hour)		HK$ 180.00/hour

[18 marks]

5. a) A new mega bridge is going to be constructed in a developing country. Given the scale and complexity of the project, two contractors formed a joint venture company specifically for this project which has subsequently won the tender of this project. Explain to the new joint venture company the key characteristics of functional, project-based and matrix organisation. [18 marks]

   b) Discuss which organisation structure would serve best for this project and prepare a simple organisation chart based on the selected form of organisation setting. [7 marks]

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